ASSOCIATION OF SCHOOLS & PROGRAMS OF PUBLIC HEALTH

THE VOICE OF ACADEMIC PUBLIC HEALTH

# PHACT – Public Health AI Curriculum Transformation Platform

*AI-powered curriculum transformation for academic public health*

## USERS

Faculty

ASPPH Member Schools & Programs

ASPPH Staff / Admins

## ACCESS CHANNELS

Web Application Responsive UI

Secure Login

APIs Future Integrations

## DATA & CONTROL FLOW LEGEND

→ Data Flow

⇢ Control / Orchestration Flow

↔ Two-Way Data Exchange

## PHACT APPLICATION LAYER

Secure Web Platform (MyASPPH)

**Curriculum Transformation Assistant**
Analyze syllabi, identify AI integration opportunities, generate recommendations

**Competency Intelligence Engine**
Map workforce needs & pedagogies to AI competencies

**Resource Repository**
AI-enabled learning modules, teaching resources & curriculum adaptations

**Faculty Learning Communities**
Collaborate, share best practices & resources

**AWS AI Sandbox Environments**
Experiment with AI tools, models & public health datasets in a safe space

## IDENTITY & SECURITY

AWS IAM Identity Center

Role Based Access Control

Amazon Cognito

AWS Key Management Service

## AI ORCHESTRATION & APPLICATION LOGIC

**Amazon Bedrock Foundation Models**
Generate, analyze, summarize & reason with retrieved institutional knowledge

**Amazon Bedrock AgentCore (Agentic Orchestration)**
Orchestrates multi-step workflows: plan, retrieve, generate, evaluate, human review & next actions

AWS Lambda / Event-driven Compute

AWS Step Functions / Workflow Orchestration

**Amazon Bedrock Guardrails**
Safety, content filters, PII protection, topic controls & responsible AI policies

**AI SANDBOX PLATFORM**
Amazon SageMaker Notebooks & Studio •AWS Bedrock Playground • Sandbox Data (Anonymized Datasets)
Safe experimentation, model exploration, training & evaluation

## RAG / KNOWLEDGE LAYER

Grounding & Retrieval

Amazon Bedrock Knowledge Bases
Semantic search, retrieval, reranking & source citations

Bedrock Knowledge Base Index / S3 Vector Embeddings

## ASPPH KNOWLEDGE & DATA SOURCES

- Workforce Job-Task Analyses
- Competency Frameworks
- Accreditation-Related Guidance
- Educational Resources & Toolkits
- ARIE & Curated AI Resources
- Faculty-Contributed Materials
- Syllabi, Assignments, Learning Objectives
- Public Health Datasets (for Sandbox)
- Amazon S3 (Secure Storage) — Raw, processed & generated content

## FACULTY OUTPUTS & IMPACT

- Curriculum Recommendations
- Competency Mappings
- Adapted Assignments & Learning Modules
- Teaching Resources
- Shared Community Resources

## HUMAN REVIEW & VALIDATION

Faculty review, validate & adapt AI-generated outputs before implementation

## APPROVED / SHARED / IMPLEMENTED

## CROSS-CUTTING SERVICES

- Amazon CloudWatch Monitoring & Logging
- AWS CloudTrail Audit & Compliance
- AWS Backup Data Protection
- AWS WAF Application Security
- Amazon Virtual Private Cloud Network Isolation
- AWS Secrets Manager Secrets & Credentials

## GOVERNANCE, COMPLIANCE & RESPONSIBLE AI

Data privacy by design • Least privilege access • Encryption in transit & at rest • Audit logging • Source attribution & transparency • Human-in-the-loop • Continuous evaluation & improvement